Fundamentals of Machine Learning
Mock Midterm (Lectures 1 to 5)

Two hours, 100 points, no documents; written in the professor's own format and at his level of difficulty, with no exercise reused from the sheets

Fundamentals of Machine Learning

2026-10-30

Central question. When a learning system must act under costs, on data it has partly never seen, with models whose assumptions can be wrong, which of its decisions follow from the mathematics and which from the choices of its designer?

Instructions. Explain your reasoning and state any assumptions. A derivation with a wrong final number earns most of the marks; a bare number earns few. Where a question says "explain", three sentences in the course's vocabulary beat a page. Suggested time per problem is given; keep ten minutes to reread. Total 100 points.

How this paper was predicted. Every tutorial, exercise and review question of the course has the same shape: a security scenario, a short derivation from a definition, numbers small enough for the hand, then "explain why" and "distinguish". The sheets were adapted from a few standard sources (Uppsala SML sessions, Murphy chapter 5, ISLR chapter 8, the XGBoost paper), so the midterm will re-skin the same mechanisms with new stories and a twist in each. The twists here are the ones he has not used yet.

Problems

1. Account takeover: from the objective to the action — 20 minutes — [18 pts]

A bank receives login events. Each event records the device identifier, the IP geolocation, the time since the previous login, the number of failed password attempts in the last hour, and whether the device was seen before on this account. A fraud team later labels each login as legitimate or takeover from chargeback disputes filed within 60 days. The deployed system must score each login and choose between letting it through, blocking it, or requiring a one-time code sent to the registered phone.

(a) Identify the physical object, the representation x, the target Y, the model output, the action set, and the learning signal. [3 pts]

(b) The data also contains "support ticket opened within 7 days" and "login at night". Classify each as legitimate information, target leakage, or a potential shortcut, and explain the difference between the last two. [3 pts]

(c) A block costs $C_{FP} = 1$ (customer friction); a missed takeover costs $C_{FN} = 30$. The one-time code costs 0.5 regardless of the class and resolves the case correctly with probability $q = 0.9$; with probability $1 - q$ the attacker passes it and the takeover is missed. Write the conditional risk of the three actions as functions of $p = \mathbb{P}(Y = 1 \mid x)$ and derive the three decision regions. [6 pts]

(d) The posterior was produced by a model trained on a balanced sample (half takeovers), while takeovers are 0.3% of deployment logins. For a login with training posterior 0.80, compute

the deployment posterior under prior shift and state the action from part (c). Explain in two sentences why skipping this correction would be worse here than in a problem with one threshold. [4 pts]

(e) State one reason the fraud labels can be wrong without the features being wrong, and what that does to the best achievable error. [2 pts]

2. The loss decides, even for a discrete target — 15 minutes — [14 pts]

For a given compromised host, the time to full compromise Y (in days) takes the values $1, 2, 5, 20$ with conditional probabilities $0.4, 0.3, 0.2, 0.1$. A model must output one number a.

(a) Find an optimal a under squared loss and under absolute loss. Say in one sentence why the first is a value Y never takes and whether that matters. [4 pts]

(b) Under-predicting costs 9 per day, over-predicting costs 1 per day. Derive the optimal a. Show that it is not unique and give the whole set of optimal predictions. [5 pts]

(c) Compute the expected loss of $a = 2$, $a = 5$ and $a = 20$ under the asymmetric loss and explain, with these numbers, why "predict the worst case" is not the cautious choice it seems. [3 pts]

(d) A deployed model outputs only $\mathbb{E}[Y \mid x]$. The security team changes the loss to the asymmetric one. Explain what the model would have to output instead and why retraining with a different loss is not the only option. [2 pts]

3. Capacity, bias and variance, and an honest protocol — 25 minutes — [20 pts]

Part I. A regression target is $Y = f(X) + \varepsilon$ with $f(x) = x^2$ and $\varepsilon \sim N(0, 1)$. At the query $x = 0$, the training inputs nearest to the query are, in order of distance, $0, \pm 0.5, \pm 1$. A k-NN regressor averages the k nearest training responses.

(a) Define bias and variance of $\hat{f}(0)$ over repeated draws of the training responses (inputs fixed). [2 pts]

(b) Compute bias, variance and expected squared error at $x = 0$ for $k = 1, 3, 5$. Which k is best here, and which term each change in k trades against which? [6 pts]

(c) Explain why the ranking of k would change if the training set were ten times denser near 0, without recomputing. [2 pts]

Part II. A team compares four intrusion detectors. It standardises the whole dataset, selects the 30 best features by correlation with the label on the whole dataset, then runs 5-fold cross-validation on the result, picks the detector with the best mean fold score, and reports that score.

(d) Name every mechanism that makes the reported score optimistic, and say for each whether it inflates the score of *every* detector or only of the selected one. [4 pts]

(e) Rewrite the protocol so that it is honest, in at most six steps. [3 pts]

(f) Two detectors have the same true error $R = 0.10$ and independent validation noise with $\sigma = 0.015$. Using $\min(a, b) = \frac{1}{2}(a + b - |a - b|)$ and $\mathbb{E}|\varepsilon_1 - \varepsilon_2| = 2\sigma/\sqrt{\pi}$, compute the expected reported error. Then explain, without computing, in which direction and why the optimism changes with 50 detectors and with a larger validation set. [3 pts]

4. Generative models: when the assumption bends — 25 minutes — [20 pts]

Part I. A single standardised feature x (log bytes per flow) has class-conditional models $p(x \mid y{=}0) = N(0, 1)$ and $p(x \mid y{=}1) = N(2, 4)$ (variance 4), with priors $\pi_0 = 0.9$, $\pi_1 = 0.1$.

(a) Starting from Bayes' rule, derive the log-odds $\log \frac{p(y=1|x)}{p(y=0|x)}$ as an explicit function of x and say why it is quadratic and not linear. [5 pts]

(b) Find the decision boundary (all points where the log-odds is zero) and describe the attack region. [4 pts]

(c) For $x = 10$, compute the posterior of class 1 and comment: both likelihoods are tiny, so what exactly is the model confident about, and what would a security analyst want the system to say instead? [3 pts]

(d) If both variances were equal, what would change in (a) and (b)? One sentence each. [2 pts]

Part II. A naive Bayes detector uses two binary features with likelihood ratio $p(x_j = 1 \mid \text{attack})/p(x_j = 1 \mid \text{benign}) = 3$ each, equal priors, and both features are on.

(e) Compute the attack posterior. Now suppose x_2 is an exact copy of x_1. What is the true posterior, what does naive Bayes report, and why can the ranking of flows still be correct while the probabilities are not? [4 pts]

(f) Give one reason to prefer logistic regression on these two features and one reason to prefer the generative model, each tied to a property of *this* data. [2 pts]

5. Ensembles: the variance floor and a leaf that disagrees with itself — 25 minutes — [18 pts]

(a) M bagged trees each have prediction variance $\sigma^2 = 1$ at a query and pairwise correlation $\rho = 0.4$. Derive the variance of their average. How many trees are needed to bring it to at most 0.45, and what is the smallest value any number of trees can reach? [5 pts]

(b) Feature subsampling lowers the correlation to $\rho = 0.2$ but raises each tree's variance to $\sigma^2 = 1.2$. For which M is the decorrelated forest better? Interpret the result in terms of "weaker trees, stronger forest". [3 pts]

(c) Gradient boosting for binary classification with the logit F and $p = \sigma(F)$. A leaf contains two examples with $y = (1, 0)$ and current $p = (0.3, 0.6)$. Compute the gradients g_i, the Hessians h_i, and the Newton leaf value for $\lambda = 0$ and $\lambda = 1$. Explain, from the signs of the g_i, why this leaf is nearly useless and what the tree would have to do about it. [5 pts]

(d) With $\lambda = 1$, compute the gain of splitting that leaf into two singletons and state the largest γ for which the split is kept. [3 pts]

(e) Out-of-bag error is free for a random forest. Explain in two sentences why there is no equivalent free estimate for a boosted ensemble. [2 pts]

6. Four claims from a product meeting — 10 minutes — [10 pts]

For each claim, say whether it is right, wrong, or right only under a condition, and justify in at most three sentences with the course's vocabulary and one number where possible.

(a) "The detector reaches 99.5% accuracy on the test set, so it is ready." Takeovers are 0.3% of logins.

(b) "We held out 20% of the data, tried 50 configurations on it, and the best one scored 0.93, so 0.93 is our test performance."

(c) "We will switch to a random forest because forests do not overfit."

(d) "We trained on a resampled 50/50 set, so we deploy with threshold 0.5."